

Lab 10: RAM and ROM

Today we will work with RAM and ROM.

Discussion – How are we doing?

1. Any issues?

Demonstration – BRAM IP

1. Using the Vivado BRAM IP in block design
2. Addressing
3. Write functionality

Continuation Lab 9 (UART)

1. If you haven't finished Lab 9, work on Lab 9

Task – RAM and ROM

1. Write both a RAM and a ROM VHDL entity
 - a. Make the word width configurable with generic parameter `word_width`
 - b. Make the memory depth configurable with generic parameter `mem_depth`
2. Test and verify in simulation

Task – RAM with Counter

1. Using your counter (set to 4 bits), your RAM (set to 4 bits width), and your debouncer, create a design which does the following:
 - a. One button increments the counter (remember debounce)
 - b. One button decrements the counter (remember debounce)
 - c. The counter points to an address in the RAM
 - d. The 4 bit data stored on the address is displayed on the 4 LEDs
 - e. One button resets everything (address pointer, RAM, etc.)
2. Program your Pynq and verify the functionality